Problem Statement: Insurance Policy Management System

Design and implement a C++ program for an Insurance Policy Management System using Object-Oriented Programming concepts.
The program must demonstrate the following concepts:
• Abstract Class
• Runtime Polymorphism (Virtual Functions and Method Overriding)
• Array of Objects
• Utility Class with Static Function

System Description
An insurance company provides different types of policies such as Life Insurance and Health Insurance. Each type of policy has its own formula for calculating the premium. The
program should calculate the premium based on the policy type.

Requirements
1. Create an Abstract Base Class InsurancePolicy
Data Members
• policyId
• policyHolderName
• sumInsured
Member Functions
• acceptPolicyDetails() – Accept policy details from the user
• displayPolicyDetails() – Display policy details
• calculatePremium() – Pure Virtual Function that must be overridden in derived classes
virtual double calculatePremium() = 0;

2. Create Derived Class LifeInsurance
Additional Data Member
• age
Premium Calculation Formula
Premium = (sumInsured × 0.02) + (age × 100)
Example:
If sumInsured = 500000 and age = 40
Premium = (500000 × 0.02) + (40 × 100)
Premium = 10000 + 4000
Premium = 14000

Override the calculatePremium(), acceptPolicyDetails (),displayPolicyDetails function to implement this formula.

3. Create Derived Class HealthInsurance
Additional Data Member
• numberOfFamilyMembers
Premium Calculation Formula
Premium = (sumInsured × 0.015) + (numberOfFamilyMembers × 2000)
Example:
If sumInsured = 400000 and familyMembers = 3
Premium = (400000 × 0.015) + (3 × 2000)
Premium = 6000 + 6000
Premium = 12000
Override the calculatePremium(),acceptPolicyDetails (),displayPolicyDetails() function accordingly.

4. Create a Utility Class PolicyUtility/PrintToScreen
This class will contain static functions used across the system.
Static Function

static void showCompanyPolicyRules(InsurancePolicy* ptr)
This function should:
Call calculatePremium() using runtime polymorphism
• Display policy details and calculated premium.

5. In main() Function
• Create an array of objects (for example: 4 policies).
• Accept policy details from the user.
• Store objects of LifeInsurance and HealthInsurance.
• Display details for all object. Call utility fucntion

Concepts Demonstrated
• Abstract Class → InsurancePolicy
• Runtime Polymorphism → Overridden calculatePremium() methods
• Array of Objects → Managing multiple policy records
• Static Utility Class → PolicyUtility::showCompanyPolicyRules()

SOLUTION:

File Location and Code

Header File

./InsurancePolicy.h

```
#pragma once

#include<iostream>;

using namespace std;

class InsurancePolicy {

private:
	int policyId;
	string policyHolderName;
	int sumInsured;

public:
	InsurancePolicy();
	InsurancePolicy(int policyId, string policyHolderName, int sumInsured);
	 virtual void acceptPolicyDetails();
	virtual void displayPolicyDetails();
	virtual double calculatePremium()=0;

	int getsumInsured();
};
```

./LifeInsurancePolicy.h

```
#pragma once

#include"InsurancePolicy.h";
class LifeInsurance : public InsurancePolicy {
```

```
private:
	int age;

public:
	LifeInsurance();
	LifeInsurance(int policyId, string policyHolderName, int sumInsured,int age);
	void acceptPolicyDetails();
	void displayPolicyDetails();
	double calculatePremium();

};
```

./HealthInsurance.h

```
#pragma once

#include"InsurancePolicy.h";

class HealthInsurance : public InsurancePolicy{

private:
	int numberOfFamilyMember;

public:
	HealthInsurance();
	HealthInsurance(int policyId, string policyHolderName, int sumInsured, int
numberOfFamilyMember);
	void acceptPolicyDetails();
	void displayPolicyDetails();
	double calculatePremium();

};
```

Resource File

./InsurancePolicy_src.cpp

```
#include"InsurancePolicy.h";

InsurancePolicy::InsurancePolicy() {
	policyId = 0;
	policyHolderName = "NA";
	sumInsured = 0;
}
InsurancePolicy::InsurancePolicy(int policyId,string policyHolderName,int sumInsured) {

	this->policyId = policyId;
	this->policyHolderName = policyHolderName;
	this->sumInsured = sumInsured;
}

void InsurancePolicy::acceptPolicyDetails() {
	cout << "\n Enter the PolicyId : ";
	cin >> policyId;
	cout << "\n Enter the policyHolderName : ";
	cin >> policyHolderName;
	cout << "\n Enter the sumInured :";
	cin >> sumInsured;
}

void InsurancePolicy::displayPolicyDetails() {
	cout << "The PolicyId is :"<< policyId<<endl;
	cout << "The PolicyHolderName is: "<<policyHolderName<<endl;
	cout << "The SumInsured is: " << sumInsured<<endl;
}

int InsurancePolicy::getsumInsured() {

	return sumInsured;
}
```

./HealthInsurance_src.cpp

```
#include"HealthInsurance.h";

HealthInsurance::HealthInsurance(){
	this->numberOfFamilyMember = 0;
}
```

```
HealthInsurance::HealthInsurance(int policyId, string policyHolderName, int sumInsured, int
numberOfFamilyMember)
:InsurancePolicy(policyId,policyHolderName,sumInsured)
{
	this->numberOfFamilyMember = numberOfFamilyMember;
}
void HealthInsurance:: acceptPolicyDetails() {
	InsurancePolicy::acceptPolicyDetails();
	cout << "\n Enter the numberOfFamilyMember :";
	cin >> numberOfFamilyMember;
}
void HealthInsurance::displayPolicyDetails() {
	InsurancePolicy::displayPolicyDetails();
	cout << "The number Of FamilyMember :" << numberOfFamilyMember << endl;
}
double HealthInsurance::calculatePremium() {
	double premium;
	premium = (getsumInsured() * 0.015) + (numberOfFamilyMember * 2000);

	return premium;
}
```

./LifeInsurance_src.cpp

```
#include"LifeInsurance.h";

LifeInsurance::LifeInsurance() {
	age = 0;
}
LifeInsurance::LifeInsurance(int policyId, string policyHolderName, int sumInsured,int age)
:InsurancePolicy(policyId,policyHolderName,sumInsured){
	this->age;
}

void LifeInsurance::acceptPolicyDetails() {
	InsurancePolicy::acceptPolicyDetails();
	cout << "\n Enter the age :";
	cin >> age;

}
```

```
void LifeInsurance::displayPolicyDetails() {
	InsurancePolicy::displayPolicyDetails();
	cout << "The age of the :" << age << endl;
}

double LifeInsurance::calculatePremium() {
	double premium;
	premium = (getsumInsured() * 0.02) + (age * 10);

	return premium;
}
```

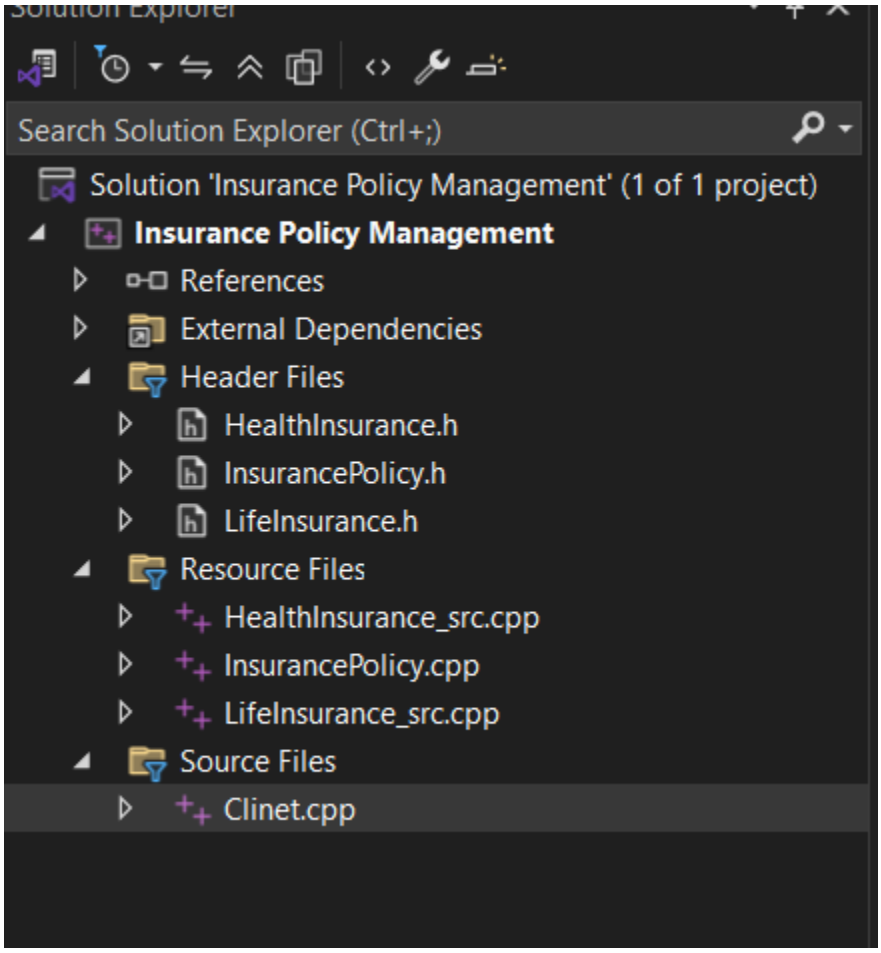
Source/Client File

./Client.cpp

```
#include "HealthInsurance.h";
#include "LifeInsurance.h";
#include"InsurancePolicy.h";

class PrintToScreen {

public:
	static void showCompanyPolicyRules(InsurancePolicy* ptr) {
		ptr->acceptPolicyDetails();

		if (typeid(*ptr) == typeid(HealthInsurance)) {
			HealthInsurance* hitr;
			hitr = dynamic_cast<HealthInsurance*>(ptr);
			cout << "HelathInsurance Calculated Premium :" <<
hitr->calculatePremium() << endl;
		}
		if (typeid(*ptr) == typeid(LifeInsurance)) {
			LifeInsurance* lptr;
			lptr = dynamic_cast<LifeInsurance*>(ptr);
			cout << "LifeInsurance Calculated Prenium : " <<
lptr->calculatePremium() << endl;
		}
		ptr->displayPolicyDetails();
```

```
        }
};

int main() {

        InsurancePolicy* policies[4];
        int choice;
        for (int i = 0;i < 4;i++) {
                cout << "1. Add HealthInsurance 2. LifeInsurance :";
                cin >> choice;

                switch (choice) {
                case 1:
                        policies[i] = new HealthInsurance();
                        PrintToScreen::showCompanyPolicyRules(policies[i]);
                         break;
                case 2:
                        policies[i] = new LifeInsurance();
                        PrintToScreen::showCompanyPolicyRules(policies[i]);
                          break;
                }

        }



}
```

STRUCTURE OF PROGRAM / FILE



OUTPUT

```
Microsoft Visual Studio Debu

Enter choice for Policy 1:
1 for HealthInsurance
2 for LifeInsurance
Choice: 1

 Enter the PolicyId : 01

 Enter the policyHolderName : Vikas

 Enter the sumInured :40000

 Enter the numberOfFamilyMember :3
HelathInsurance Calculated Premium :12000
The PolicyId is :1
The PolicyHolderName is: Vikas
The SumInsured is: 40000
The number Of FamilyMember :3

Enter choice for Policy 2:
1 for HealthInsurance
2 for LifeInsurance
Choice: 1

 Enter the PolicyId : 02

 Enter the policyHolderName : Vivek

 Enter the sumInured :40000

 Enter the numberOfFamilyMember :5
HelathInsurance Calculated Premium :16000
The PolicyId is :2
The PolicyHolderName is: Vivek
The SumInsured is: 40000
The number Of FamilyMember :5
```

```
Enter choice for Policy 3:
1 for HealthInsurance
2 for LifeInsurance
Choice: 2

 Enter the PolicyId : 03

 Enter the policyHolderName : Anuj

 Enter the sumInured :500000

 Enter the age :2
LifeInsurance Calculated Prenium : 10020
The PolicyId is :3
The PolicyHolderName is: Anuj
The SumInsured is: 500000
The age of the :2

Enter choice for Policy 4:
1 for HealthInsurance
2 for LifeInsurance
Choice: 2

 Enter the PolicyId : 04

 Enter the policyHolderName : Vivansh

 Enter the sumInured :60000

 Enter the age :26
LifeInsurance Calculated Prenium : 1460
The PolicyId is :4
The PolicyHolderName is: Vivansh
The SumInsured is: 60000
The age of the :26

D:\c++ pratice\Day7\Insurance Policy Management\x64\Debug\Insurance Policy Management.exe (process 238
64) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatical
ly close the console when debugging stops.
Press any key to close this window . . .
```